A Literature-Implied Partial-Subcase Answer for the Schreier Self-Sphere Case

Run `fa_banach_001`, packet `2110.11786_schreier_self_tingley_fakhoury`

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Packet Status

This packet records a literature-implied answer to a partial subcase, not an original proof from the run. The source problem in Antunes–Beanland asks for the full Mazur–Ulam property for combinatorial Banach spaces, including arbitrary target spaces. Fakhoury's later theorem proves the real self-sphere Schreier case. Thus the result below should be treated as a provenance/status packet for a solved subcase and not as a full answer to Problem 3.2.

Source Problem

The source is Leandro Antunes and Kevin Beanland, *Surjective isometries on Banach sequence spaces: a survey*, arXiv:2110.11786. Problem 3.2 asks:

> Let $\mathcal{F}$ be a regular family and X a Banach space. Suppose that $V : S_{X_\mathcal{F}} \to S_X$ is a surjective, not necessarily linear, isometry. Does there exist a linear surjective isometry $U : X_\mathcal{F} \to X$ so that $U|_{S_{X_\mathcal{F}}} = V$?

The same page states that this problem is open even when $\mathcal{F} = \mathcal{S}_1$, the classical Schreier family.

Problem 3.1. *For which combinatorial Banach spaces can we explicitly describe the group of surjective isometries?*

The following linear extension problem restricted to combinatorial Banach spaces is open.

Problem 3.2. *Let $\mathcal{F}$ be a regular family and X and Banach space. Suppose that V : $S_{X_\mathcal{F}} \to S_X$ is a surjective (not necessarily linear) isometry. Does there exist a linear surjective isometry $U : X_\mathcal{F} \to X$ so the $U|_{S_\mathcal{F}} = V$?*

This problem is open even in the case $\mathcal{F} = \mathcal{S}_1$. Let us make a few remarks on this problem. A space having the desired property is said to have the Mazur-Ulam property. Famously, Mazur and Ulam proved that a surjective isometry between Banach spaces that fixes 0 is a linear isometry. There is a vast literature verifying that various spaces have the Mazur-Ulam property [37]. In particular, all classical sequence spaces have the property [31, 32, 42]. As for non-classical sequence spaces, Tan showed that surjective isometries on spheres of Tsirelson's space [60] and James' space (in one of the natural norms) [61]

Supporting Literature

The supporting source is Micheline Fakhoury, *Tingley's problem for Schreier spaces and their p-convexifications*, arXiv:2412.00891. Its Theorem 1.1 states that if $(\mathcal{S}_\alpha)_{\alpha<\omega_1}$ is a good transfinite Schreier sequence, $p \in [1, \infty)$, and $1 \le \alpha < \omega_1$, then every surjective isometry of the unit sphere of $X_{\mathcal{S}_\alpha,p}$ extends to a linear surjective isometry of $X_{\mathcal{S}_\alpha,p}$; moreover, for $p \in \{1, 2\}$ the goodness assumption is not needed.

is made up of successor ordinals, $\alpha_n = \beta_n + 1$, and the sequence $(\mathcal{S}_{\beta_n})_{n\in\mathbb{N}}$ is increasing.

Note that there exist transfinite Schreier sequences that are good by [7, Proposition 3.1 (viii)] (see also [9, Fact 3.1]).

We are going to prove the following theorem.

Theorem 1.1. *Let $(\mathcal{S}_\alpha)_{\alpha<\omega_1}$ be a good transfinite Schreier sequence and let $p \in [1, \infty)$ and $1 \leqslant \alpha < \omega_1$. Then every surjective isometry of the unit sphere of $X_{\mathcal{S}_\alpha,p}$ can be extended to a linear surjective isometry of $X_{\mathcal{S}_\alpha,p}$. Moreover, if $p \in \{1, 2\}$, it is not necessary to assume that $(\mathcal{S}_\alpha)_{\alpha<\omega_1}$ is good.*

In the sequel, we fix a transfinite Schreier sequence $(\mathcal{S}_\alpha)_{\alpha<\omega_1}$, an ordinal $1 \leqslant \alpha < \omega_1$ and $p \in [1, \infty)$. Moreover, we assume that $(\mathcal{S}_\alpha)_{\alpha<\omega_1}$ is good when $p \in (1, \infty) \setminus \{2\}$.

We denote by $\mathbb{S}$ the unit sphere of $X_{\mathcal{S}_\alpha,p}$ and we fix a surjective isometry $T : \mathbb{S} \to \mathbb{S}$. Our aim is to show that T can be extended to a linear surjective isometry $\tilde{T} : X_{\mathcal{S}_\alpha,p} \to X_{\mathcal{S}_\alpha,p}$.

Fortunately, we know by [2], [5] (for $p = 1$) and [9] (for $p > 1$) that any linear surjective isometry J of $X_{\mathcal{S}_\alpha,p}$ is *diagonal*, *i.e.* there exists a sequence of signs $(\theta_i)_{i\in\mathbb{N}}$ (*i.e.* $\theta_i = \pm 1$) such that $Je_i = \theta_i e_i$ for all $i \in \mathbb{N}$. It follows that if T can be extended to a linear surjective isometry of $X_{\mathcal{S}_\alpha,p}$, then there exists a sequence of signs (θ_i) such that $T(x)(i) = \theta_i x(i)$ for any $x \in \mathbb{S}$ and $i \in \mathbb{N}$. Conversely, this formula does define a linear surjective isometry of $X_{\mathcal{S}_\alpha,p}$. Therefore, our aim is to prove the following theorem.

Theorem 1.2. *There exists a sequence of signs $(\theta_i)_{i\in\mathbb{N}}$ such that for any $x \in \mathbb{S}$ and $i \in \mathbb{N}$, we have $T(x)(i) = \theta_i x(i)$.*

If we are able to prove Theorem 1.2, then we immediately obtain Theorem 1.1. The main step to prove it is to show the following proposition.

Proposition 1.3. *There exists a sequence of signs $(\theta_i)_{i\in\mathbb{N}}$ such that $T(e_i) = \theta_i e_i$ for any $i \in \mathbb{N}$.*

Note that Proposition 1.3 re-proves some theorems from [2], [5] and [9] concerning the description of linear surjective isometries of $X_{\mathcal{S}_\alpha,p}$, for $1 \leqslant p < \infty$, and generalizes [9, Proposition 3.2 (b)] (in the real case) because we do not need to assume that $(\mathcal{S}_\alpha)_{\alpha<\omega_1}$ is good when $p = 2$.

Idea of the Proof

The identification is direct. The source problem singles out the classical Schreier space $X_{\mathcal{S}_1}$ as an unsolved case. Fakhoury's notation satisfies $X_{\mathcal{F},1} = X_{\mathcal{F}}$, and $\mathcal{S}_1$ is exactly the classical Schreier family. Applying Fakhoury's theorem with $\alpha = 1$ and $p = 1$ therefore gives extension of every surjective isometry from $S_{X_{\mathcal{S}_1}}$ onto itself. The result is only a self-map subcase because Fakhoury's theorem does not allow an arbitrary target sphere S_X.

Formal Statement

Theorem. *Let $X_{\mathcal{S}_1}$ be the real Schreier space. Every surjective isometry $T : S_{X_{\mathcal{S}_1}} \to S_{X_{\mathcal{S}_1}}$ extends to a linear surjective isometry $\widetilde{T} : X_{\mathcal{S}_1} \to X_{\mathcal{S}_1}$. Consequently, the self-map real Schreier-space subcase of Problem 3.2 in arXiv:2110.11786 has an affirmative answer.*

Proof. Fakhoury defines, for a combinatorial family $\mathcal{F}$ and $p \in (1, \infty)$, the p-convexification $X_{\mathcal{F},p}$, and sets $X_{\mathcal{F},1} = X_{\mathcal{F}}$. Her introduction identifies $\mathcal{S}_1$ as the classical Schreier family and $X_{\mathcal{S}_1}$ as the Schreier space. Theorem 1.1 of arXiv:2412.00891 applies to every $1 \leq \alpha < \omega_1$ and every $p \in [1, \infty)$; when $p = 1$, no additional "good" hypothesis on the transfinite Schreier sequence is needed.

Specialize that theorem to $\alpha = 1$ and $p = 1$. Then every surjective isometry of the unit sphere of $X_{\mathcal{S}_1,1} = X_{\mathcal{S}_1}$ extends to a linear surjective isometry of $X_{\mathcal{S}_1,1} = X_{\mathcal{S}_1}$. This is precisely the subcase of Antunes–Beanland Problem 3.2 in which $\mathcal{F} = \mathcal{S}_1$ and the target space is again $X_{\mathcal{S}_1}$. $\square$

Limitations

This packet does not prove that $X_{\mathcal{S}_1}$ has the full Mazur–Ulam property as formulated in the source survey, because the source asks about surjective isometries $S_{X_{\mathcal{S}_1}} \to S_X$ for arbitrary Banach target spaces X. It also does not address complex Schreier spaces. The result recorded here is the real self-sphere case only.

Literature And Novelty Check

Before creating this packet, the run's lightweight indexes were searched for arXiv:2110.11786, the survey title, "Banach sequence spaces", "Schreier", and Tingley/Mazur–Ulam keywords; no existing packet, attempt, or proof-gap entry covered this target. A bounded web/arXiv search on June 14, 2026 using queries including "Tingley's problem Schreier space", "Schreier space Mazur-Ulam property", and "surjective isometries sphere Schreier space" identified Fakhoury's arXiv:2412.00891 paper.

Human Review Recommendation

Review as a literature-implied partial-subcase answer. The main checks are the scope distinction between self-map sphere isometries and the full arbitrary target Mazur–Ulam property, and the notational identification $X_{\mathcal{S}_1,1} = X_{\mathcal{S}_1}$.

References

[1] L. Antunes and K. Beanland, *Surjective isometries on Banach sequence spaces: a survey*, arXiv:2110.11786.

[2] M. Fakhoury, *Tingley's problem for Schreier spaces and their p-convexifications*, arXiv:2412.00891.